COMSATS UNIVERSITY ISLAMABAD
WAH CAMPUS
Department of Computer Science

Semester Project Proposal
Course: CSC462 — Artificial Intelligence

Safar AI

Pakistan Smart Route Predictor

Submitted by:
Hamza Ali
BS Software Engineering
Session: 2024-2028
FA24-BSE-124/WAH

TEAM MEMBERS:
SHAHZAIB SHOAIB RATHORE
ADNAN SHAH
COMSATS University Islamabad, Wah Campus

1. Introduction

Safar AI is a Python-based intelligent route planning and road blockage prediction system built specifically for Pakistan. The system uses classical AI algorithms, machine learning, and rule-based reasoning to help travelers plan intercity journeys with full awareness of road blockages, fuel costs, travel time, and route safety.

Pakistan's road network faces unique disruptions — political protests, VVIP movements, floods, and landslides — that existing navigation tools like Google Maps and Waze do not account for. Safar AI fills this gap by providing route intelligence that is specifically designed for Pakistani roads and travelers.

2. Problem Statement

Intercity travelers in Pakistan face the following challenges that no existing tool addresses:

- Road closures due to political protests and sit-ins blocking major highways
- VVIP movements causing sudden, unannounced road closures in cities
- No tool calculates fuel cost in PKR based on vehicle type and live fuel rates
- No safety scoring system exists for Pakistani road segments
- Travelers have no single platform to plan long routes with rest areas and fuel stops

3. Objectives

- Build a real-road routing engine using A* algorithm on OpenStreetMap data for Pakistan
- Develop a rule-based expert system to detect and classify road blockages
- Train a machine learning classifier to predict blockage probability on route segments
- Calculate fuel cost in PKR for petrol, diesel, and CNG vehicles
- Build a conversational agent (Raasta) that understands Urdu, Roman Urdu, and English
- Display everything on a real interactive Pakistan map in a single-page web application
- Store all user queries and system responses in a database for analysis

4. Alignment with CSC462 AI Course

Every core component of Safar AI directly maps to a topic covered in CSC462 Artificial Intelligence:

Course Unit	Topic	How Used in Safar AI
Unit 1	AI Agents & Rationality	Raasta is modelled as a rational AI Agent with PEAS definition
Unit 2	A* Search Algorithm	Core routing engine — finds optimal path on OSM road graph
Unit 5	Expert Systems & Chaining	Rule-based blockage detection using forward chaining on news data
Unit 6	Machine Learning	Random Forest classifier predicts blockage probability per route

5. System Modules

Module A — Predictive Road Blockage Engine

Collects road blockage data from news RSS feeds and a manually maintained dataset. A rule-based expert system classifies blockages by type, location, and severity. A trained ML classifier predicts the probability of blockage on each segment of the selected route.

Module B — Smart Routing and Cost Engine

Computes optimal routes using A* on a real OpenStreetMap road graph of Pakistan downloaded via osmnx. Applies a fuel cost layer using PKR petrol, diesel, and CNG rates. Calculates a safety score and ETA per route. Highlights fuel stations and rest areas along the way.

Module C — Conversational Agent (Raasta)

Accepts text input in Urdu, Roman Urdu, or English. Detects language, extracts source city, destination city, vehicle type, and fuel type. Calls all backend engines and returns a complete route summary in the same language the user typed in.

Module D — Map and Visualization

Renders a real-world Pakistan map using Folium on OpenStreetMap tiles. Draws the computed route as a polyline on actual roads. Shows blockage markers, fuel station pins, rest area locations, and a heatmap of high-risk zones.

6. Technology Stack

Category	Tool	Purpose
Frontend	Streamlit	Single-page web application
Map	Folium + OpenStreetMap	Real interactive Pakistan map
Road Graph	osmnx + networkx	Download and query OSM road network
Routing	A* Algorithm	Optimal path computation
ML	scikit-learn	Blockage probability classifier
NLP / Agent	langdetect + fuzzywuzzy	Language detection, city matching
Charts	Plotly	Cost, ETA, safety visualizations
Database	SQLite	Store all user queries and answers
Data Scraping	requests + BeautifulSoup	Fetch blockage news from web
Deployment	Streamlit Community Cloud	Free public deployment

7. Data Sources

- OpenStreetMap — Real Pakistan road network (free, via osmnx)
- OGRA Website — Petrol, diesel, CNG prices in PKR (manual update)
- Dawn / Geo / ARY News RSS Feeds — Blockage and road closure news (free)
- OSM Overpass API — Fuel stations, rest areas, hotels along routes (free)
- Manual Dataset — Pakistan cities with coordinates, safety index, blockage history

8. Development Plan

Days	Phase	What Gets Done
1	Setup & Data	Project structure, GitHub, JSON data files, requirements installed
2-3	Routing Engine	A* on OSM graph, path computation, ETA, cost calculator
4	Blockage Engine	Rule-based expert system, blockage data, safety scoring
5	ML Model	Feature engineering, Random Forest training, prediction module
6-7	Raasta Agent	Language detection, intent extraction, query-to-answer pipeline

8	Map Layer	Folium map, route line, blockage markers, heatmap, POI pins
9	Database	SQLite schema, query logging, history display
9	Streamlit UI	Single page app, map, chat input, charts, full integration
10	Testing	End-to-end testing, bug fixes, edge cases
11	Deployment	Streamlit Cloud deployment, live URL, GitHub push
12	Documentation	README, architecture diagram, flowchart, final cleanup

9. Expected Output

- A fully working web application deployed on Streamlit Community Cloud
- Real A* routing on Pakistan OSM road network across 50+ cities
- Trained ML blockage prediction model with accuracy metrics
- Conversational agent handling Urdu, Roman Urdu, and English queries
- SQLite database logging all user interactions
- Interactive real-world Pakistan map with route, blockages, and POIs
- Clean GitHub repository with modular, documented code

10. Conclusion

Safar AI is a practical, original, and technically grounded project that directly applies concepts from CSC462 Artificial Intelligence to a real-world problem specific to Pakistan. The project uses A* search, rule-based expert systems, and machine learning — exactly as taught in the course — to build something that provides genuine value to Pakistani travelers.

The entire system is built using free and open-source tools, making it fully reproducible with zero budget. The modular structure ensures the project can be extended and improved in the future as needed.

Student Declaration

I declare that this project proposal is my own original work, submitted as part of the semester project requirement for CSC462 Artificial Intelligence at COMSATS University Islamabad, Wah Campus.

Name: Hamza Ali   Date: 02/05/2026